

SMT. TANUBEN & DR. MANUBHAI TRIVEDI COLLEGE OF INFORMATION
SCIENCE, SURAT
BACHELOR OF COMPUTER APPLICATIONS
T.Y.B.C.A. - 6TH SEM: 2024-25
SUBMISSION OF PROJECT TITLE

Group No: 27

Date: 09/02/2025

PROJECT INFORMATION:

Project Title: AgriVision - IoT Based Smart Agriculture Monitoring System
Front End Tool: React.js, HTML, CSS, Bootstrap, JavaScript
Back End Tool: Firebase (Realtime Database & Authentication)
Database: Firebase Firestore / Realtime Database
Operating System Used: Windows
Miscellaneous Tool (if any): ESP32, Arduino IDE, IoT Sensors (Soil Moisture, DHT11, LDR)
Team Size: 3

BRIEF DESCRIPTION OF PROJECT:

- Our project AgriVision is an IoT-based smart agriculture monitoring and advisory system that integrates real-time sensor data with a web platform to help farmers monitor their field conditions effectively. The system collects live data from soil moisture sensors, temperature & humidity sensors, and light intensity sensors using ESP32 microcontroller.
- The web dashboard displays real-time field data in simple language with color-coded status (Green=Normal, Yellow=Warning, Red=Critical). It provides personalized alerts and crop recommendations based on actual sensor readings, making it easy for small-scale farmers to understand their field conditions without technical knowledge.
- **Key Features:**
 - I. Real-time IoT sensor data monitoring
 - II. Smart alerts and crop recommendations
 - III. Historical data analysis with graphs
 - IV. Low-cost system (₹1,300 hardware cost)

USER-WISE FUNCTIONALITIES:**❖ Admin**

- Login
- Manage Farmers
- Manage Sensors
- Set Threshold Values (Soil, Temperature, Humidity)
- View System Analytics
- Monitor Sensor Health
- Change Password
- Log Out

❖ Farmer

- Login
- Register
- View Live Sensor Data (Soil, Temperature, Humidity, Light)
- View Alerts & Notifications
- View Smart Recommendations
- View Field Monitoring Graphs
- View Data History
- Download Data
- Change Password
- Log Out

❖ IoT Device (ESP32)

- Collect Sensor Data
- Send Data to Firebase Cloud
- Continuous Data Updates
- Sensor Status Monitoring

STUDENT INFORMATION:

No.	Name of the Student	Email Id	Contact No	Sign
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